

Social Network Analysis

Ch3: Network Models II Community detection

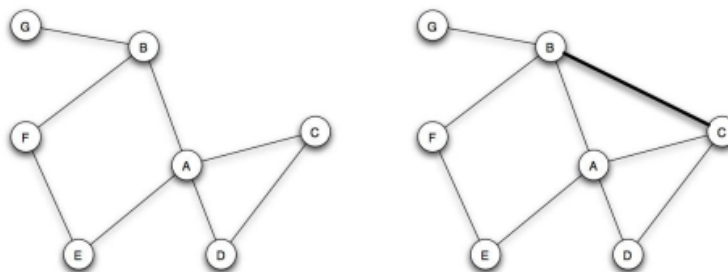
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Some slides are excerpted from this lecture: <https://ocw.mit.edu/courses/1-022-introduction-to-network-models-fall-2018/pages/lecture-notes/>

Triadic closure

- ▶ Networks are rarely static structures $\Rightarrow$ Think about their evolution
 - $\Rightarrow$ How are edges formed? $\Rightarrow$ Universal feature $\Rightarrow$ **Triadic closure**
- ▶ *If two people in a social network have a friend in common, then there is an increased likelihood that they will become friends at some point in the future*



(a) Before B-C edge forms.

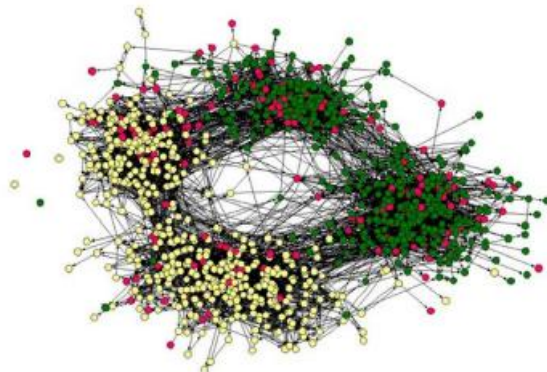
(b) After B-C edge forms.

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- ▶ Triadic closure is very natural $\Rightarrow$ Some reasons ...
 - $\Rightarrow$ **Opportunity**: B and C have a higher chance of meeting
 - $\Rightarrow$ **Trusting**: B and C are predisposed to trusting each other
 - $\Rightarrow$ **Incentive**: A might have incentive to make B and C friends

Homophily

- ▶ We tend to be **similar to our friends** ⇒ Well known for long time
 - ⇒ Age, race, interests, beliefs, opinions, affluence, ...
- ▶ **Contextual** (as opposed to **intrinsic**) effect on network formation
 - ⇒ **Contextual**: Friends because we attend the same school
 - ⇒ **Intrinsic**: Friends because a common friend introduces us

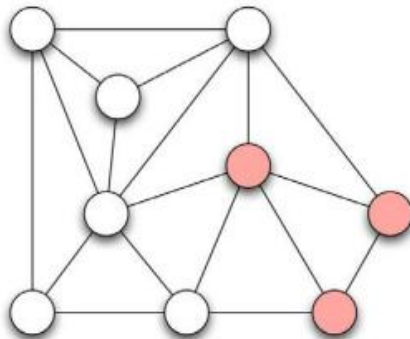


Moody, James. "Race, School Integration, and Friendship Segregation in America." *American Journal of Sociology* 107 (2001): 679–716. © University of Chicago Press. All rights reserved. This content is excluded from our Creative Commons license. For more information, see <https://ocw.mit.edu/help/faq-fair-use/>.

- ▶ In previous slide, B and C high chance of becoming friends
 - ⇒ Even if they are not aware of common knowledge of A

Measuring homophily

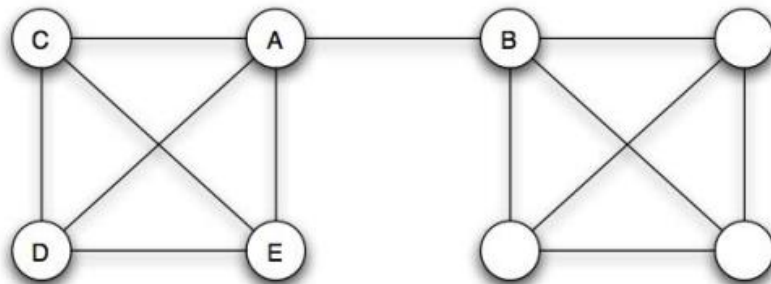
- ▶ Is homophily present or is it an **artifact** of how the network is drawn?
 - ⇒ We need to formulate a **precise mathematical measure**
- ▶ Consider a small network of girls ($q = 3/9$) and boys ($p = 6/9$)



- ▶ If edges are agnostic to gender, portion of cross-gender edges is $2pq$
 - ⇒ **Homophily Test**: If the fraction of cross-gender edges is significantly less than $2pq$, then there is evidence for homophily
 - ⇒ Cross-gender edges $5/18 < 8/18 = 2pq$ ⇒ Mild homophily

Hearing about a new job

- ▶ Mark Granovetter (1973) interviewed people that changed jobs
- ▶ Most heard about new jobs from acquaintances rather than close friends
 - ⇒ Explanation takes into account local properties and global structure

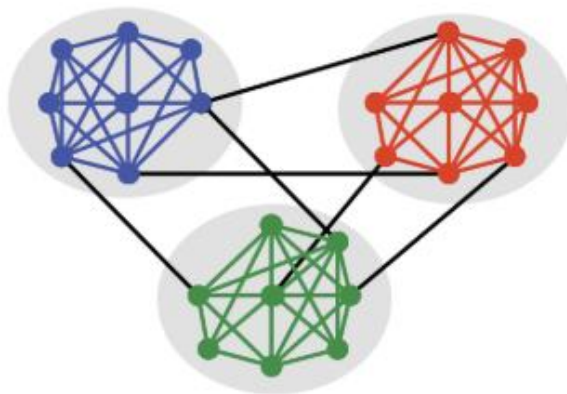


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- ▶ A's friends E, C, and D form a tightly-knit group
- ▶ B reaches to a different part of the network ⇒ New information
- ▶ Deleting (A, B) disconnects the network ⇒ (A, B) is a bridge
 - ⇒ But bridges are rare in real-world networks

Community detection and spectral clustering

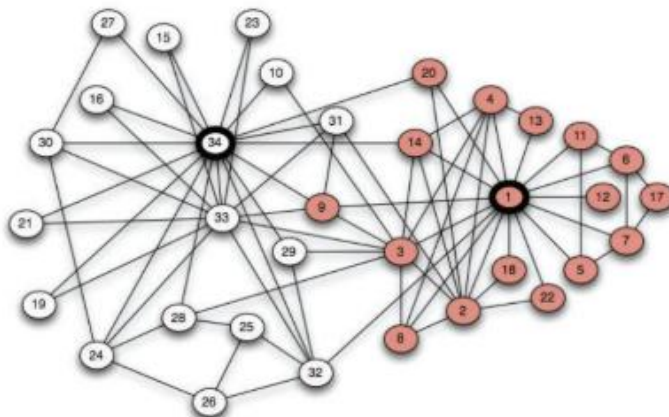
- ▶ Nodes in many real-world networks organize into **communities**
Ex: families, clubs, political organizations, urban areas, ...
- ▶ Supported by the **strength of weak ties** theory



- ▶ Community (a.k.a. group, cluster, module) members are:
 - ⇒ **Well connected** among themselves
 - ⇒ Relatively **well separated** from the rest

Zachary's karate club

- Social interactions among members of a karate club in the 70s
⇒ Canonical network for community detection methods

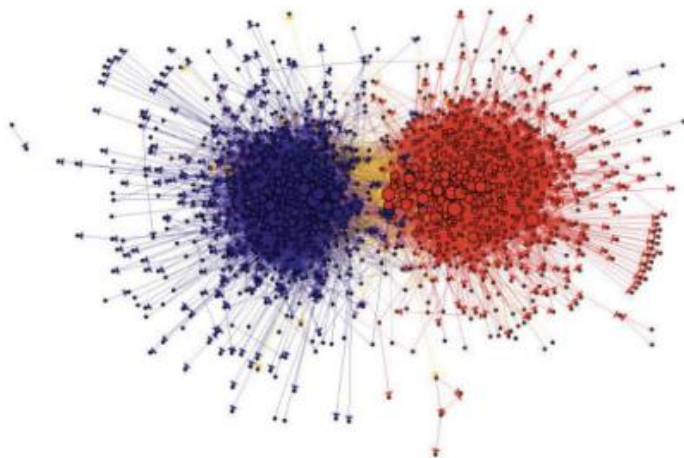


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- The club split into two during the study (white and red groups)
⇒ Offers ground-truth community membership
- Could we have predicted the split only from the network structure?

Political blogs

- The political blogosphere for the US 2004 presidential election

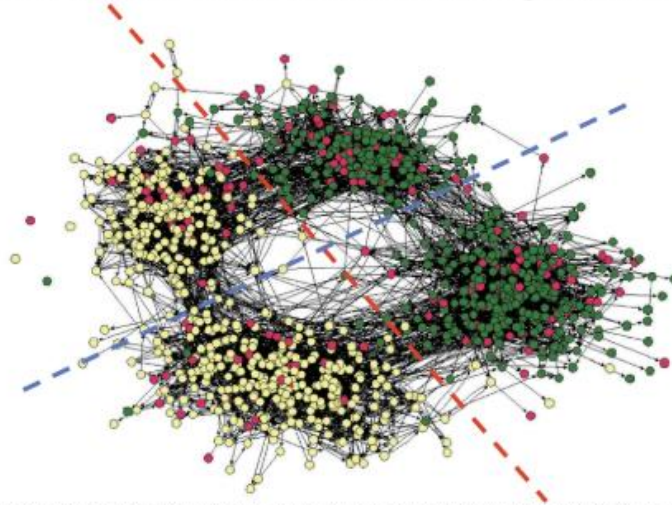


Adamic, Lada and Natalie Glance, "The Political Blogosphere and the 2004 U.S. Election: Divided They Blog," March 4, 2005. © Lada Adamic and Natalie Glance. All rights reserved. This content is excluded from our Creative Commons license. For more information, see <https://ocw.mit.edu/help/faq-fair-use/>.

- Community structure of liberal and conservative blogs is apparent
 - ⇒ Strong evidence of partisan homophily in the network
- Can we detect both parties without looking at the blogs' content?

School students

- ▶ Social network from a town's middle and high school students

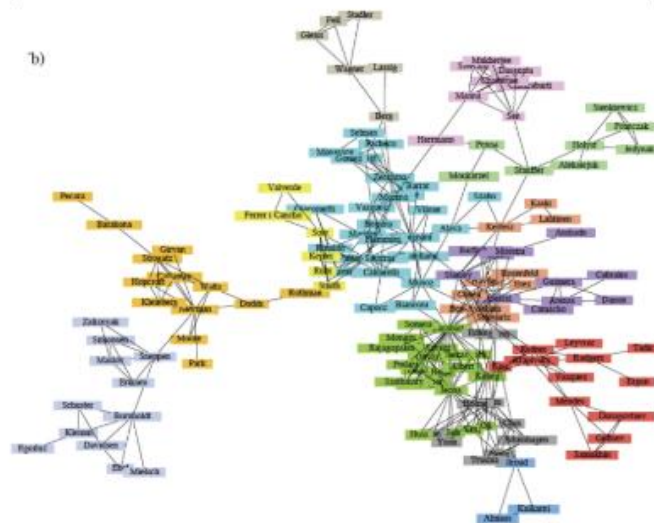


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- ▶ Two binary divisions are apparent from the structure of the network
 - ⇒ Racial division marked in red
 - ⇒ Age division (middle - high) marked in blue
- ▶ Can we estimate race and age of a student from the structure?

Physicists working on Network Science

- Co-authorship network of physicists working on networks
 - ⇒ Edges represent the existence of a collaborative publication

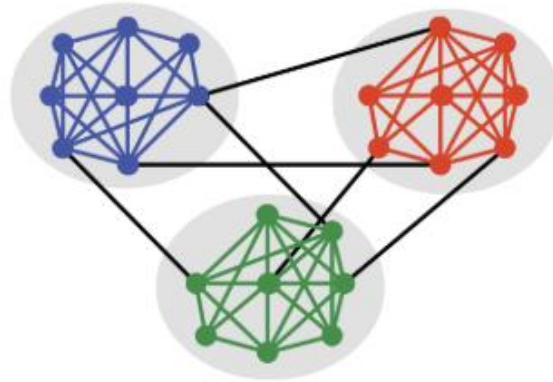


Newman, M. E. J., and M. Girvan. "Finding and evaluating community structure in networks." *Physical Review E* 69 (2004): 026113. © American Physical Society. All rights reserved. This content is excluded from our Creative Commons license. For more information, see <https://ocw.mit.edu/help/faq-fair-use/>.

- Tightly-knit subgroups are evident from the network structure
 - ⇒ Some researchers work at the boundary between two groups?
- Can we recover this information without relying on visual inspection?

Automatic detection of communities

- ▶ Recurring theme in all of the examples provided
 - ⇒ How can we automatically detect communities in a network?
- ▶ But ... what is a sensible definition of community?
 - ⇒ Multiple definitions lead to multiple community detection methods

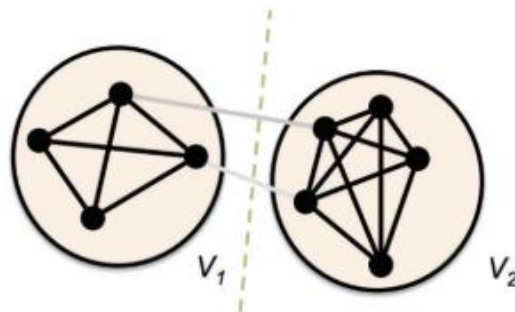


Graph partitioning and community detection

- ▶ **Community detection is a challenging problem**
 - ⇒ No universal definition of community
 - ⇒ No prior knowledge of community number or sizes
 - ⇒ Rare ground-truth data for validation
- ▶ We begin with a simpler problem ⇒ **Graph partitioning**
- ▶ Divide V into a **given number** of non-overlapping groups of a **given size**
- ▶ Graph partitioning is still a hard problem
 - ⇒ Even graph bisection (two groups, equal size) has $\binom{|V|}{|V|/2}$ possibilities
- ▶ **Exhaustive search intractable beyond small datasets**
- ▶ Need to rely on tractable relaxations of natural partitioning criteria

Graph partitioning and minimum cuts

- ▶ Community members should be well-connected among themselves
⇒ Loosely connected with members of other communities



- ▶ A **cut** C is the weight of edges between blocks V_1 and $V_2 = V \setminus V_1$

$$C = \text{cut}(V_1, V_2) = \sum_{i \in V_1, j \in V_2} A_{ij}$$

- ▶ Find cut that achieves the desired sizes in V_1 and V_2 while minimizing C